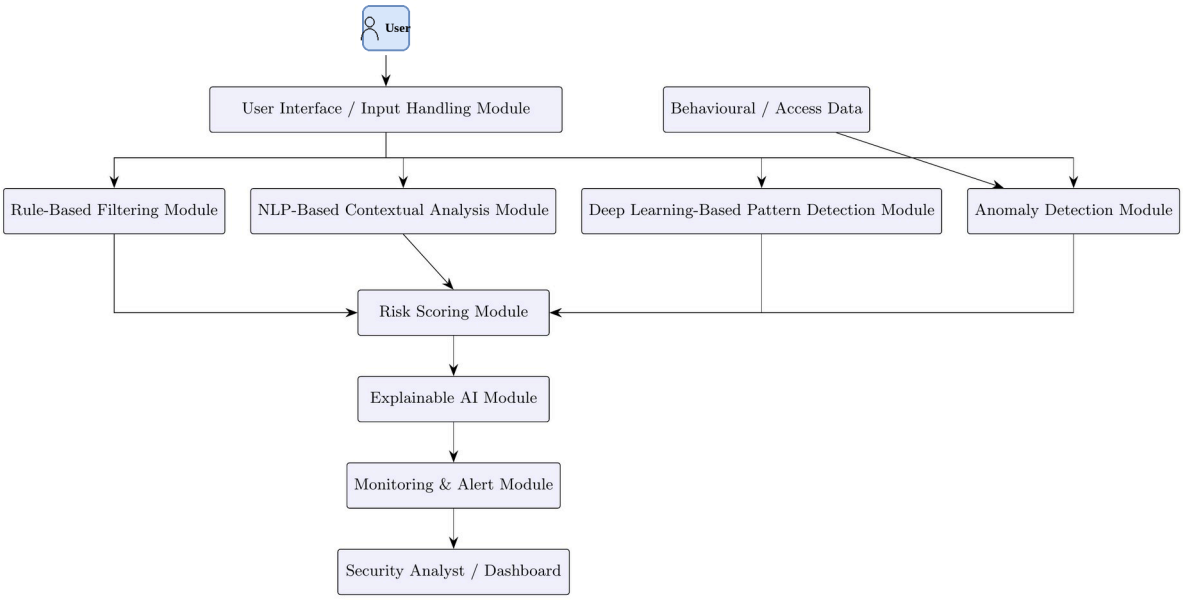



Fig. 5.1: System Architecture of the Intelligent Personal Data Leakage Detection System

## 5.2 Data Flow Diagram (DFD)

This section illustrates the data flow diagrams of the proposed system, showing the complete movement of data throughout the personal data leakage detection process, from the context-level view down to the internal steps of its more complex processes.

### 5.2.1 DFD Level 0

rated
input
eline
utput

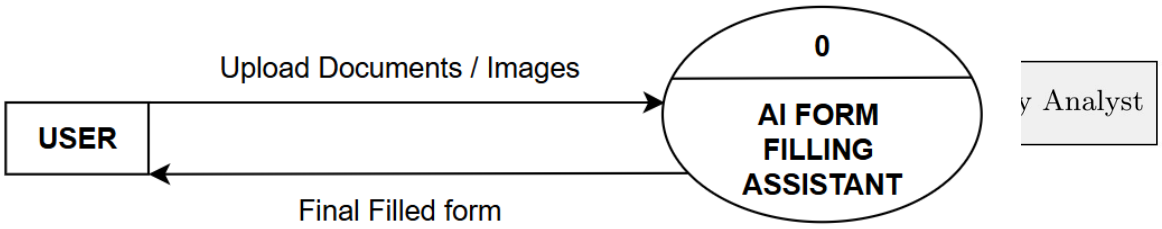



Fig. 5.2: DFD Level 0 of the Intelligent Personal Data Leakage Detection System

### 5.2.2 DFD Level 1

At Level 1, shown in Figure 5.3, the internal processing of the system is broken down further. The process begins when the input is received and validated by Process 1.0 (Input Validation). The validated input is then passed to Process 2.0 (Rule-Based Filtering), which draws on the calibration dataset (D1) for its configured patterns, and